

Algebra First-Year Exam, January 2017, Part 2

Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Let R be a nonzero commutative ring with identity. Show that R has a minimal prime ideal, i.e. a prime ideal not containing any other prime ideal.
2. Which of the following polynomials are irreducible? Explain.
 - (a) $X^3 + X + 1$ in $\mathbb{F}_2[X]$;
 - (b) $X^3 + X + 1$ in $\mathbb{Q}[X]$;
 - (c) $X^5 + 6X^2 + 9X + 3$ in $\mathbb{Q}[X]$;
3. Show that a nonzero ring R with identity (not necessarily commutative) is a division ring if and only if its only left ideals are 0 and R .
4. Let R be an integral domain.
 - (i) Define what it means for an element of R to be irreducible.
 - (ii) Define what it means for an element of R to be prime.
 - (iii) Show that a prime element is irreducible.
5. Suppose F is a field and V is an F -vector space. Denote by V' the dual of V , i.e. the F -vector space of linear maps $V \rightarrow F$.
 - (i) Show that if $\dim_F V < \infty$, then $V \simeq V'$.
 - (ii) Show that if V has infinite dimension, then $V \not\simeq V'$.
6. Show that if L/K and E/L are algebraic extensions of fields, then so is E/K .
7. Show that $GL_3(\mathbb{F}_2)$ has two conjugacy classes of elements of order 7, and find representatives of each.